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Practice exam 1

Units 3 & 4 Paper 1

Perusal time: 5 minutes

Working time: 90 minutes

Section 1: Multiple choice

Instructions

- Choose the best answer for Questions 1–20.
- This section has **20 questions** and is worth **20 marks**.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

	A	B	C	D
1.	☐	☐	☐	☐
2.	☐	☐	☐	☐
3.	☐	☐	☐	☐
4.	☐	☐	☐	☐
5.	☐	☐	☐	☐
6.	☐	☐	☐	☐
7.	☐	☐	☐	☐
8.	☐	☐	☐	☐
9.	☐	☐	☐	☐
10.	☐	☐	☐	☐
11.	☐	☐	☐	☐
12.	☐	☐	☐	☐
13.	☐	☐	☐	☐
14.	☐	☐	☐	☐
15.	☐	☐	☐	☐
16.	☐	☐	☐	☐
17.	☐	☐	☐	☐
18.	☐	☐	☐	☐
19.	☐	☐	☐	☐
20.	☐	☐	☐	☐

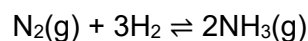
Name: Class: OXFORD
UNIVERSITY PRESS**QUESTION 1**

In a dynamic equilibrium

- A** the rate of the forward reaction is greater than the reverse reaction.
- B** bonds are being formed and re-formed constantly.
- C** the reaction is complete.
- D** the reaction consists of only products.

QUESTION 2

When the concentration of hydrogen is increased in the following reaction



the equilibrium will adjust by moving to the side of the

- A** reactants to increase the concentration of hydrogen.
- B** products to decrease the concentration of hydrogen.
- C** products to increase the concentration of hydrogen.
- D** reactants to decrease the concentration of hydrogen.

QUESTION 3

The pH of a solution of potassium hydroxide is 10. The hydroxide ion concentration in this solution is

- A** 10^{-10} M
- B** 10^{-7} M
- C** 10^{-4} M
- D** 10^{-1} M

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QUESTION 4

Iron is one of several transition metals that can exist in different oxidation states. The oxidation state of iron in $\text{Fe}(\text{OH})_3$ and FeO_4^{2-} , respectively, is

- A +3, +6
- B +3, -2
- C -3, -6
- D 0, -2

QUESTION 5

Two commonly used acid–base indicators are bromocresol green and alizarin yellow.

Indicator	pH range	Colour of acidic form	Colour of basic form
Bromocresol green	4–5.6	yellow	blue
Alizarin yellow	10–12	yellow	red

A very weak acid is titrated with sodium hydroxide. If the pH at the endpoint of the titration is approximately 10, the most suitable indicator for this titration is

- A bromocresol green and its colour will change from yellow to blue.
- B alizarin yellow and its colour will change from yellow to red.
- C bromocresol green and its colour will change from blue to yellow.
- D alizarin yellow and its colour will change from red to yellow.

QUESTION 6

Acid HX has an acid dissociation constant of 2.8×10^{-7} M. Which of the following statements is correct?

- A Acid HX is a strong acid.
- B There is significantly more H^+ than HX in an aqueous solution.
- C Acid HX completely ionises in an aqueous solution.
- D The concentration of H^+ is significantly less than the concentration of HX in solution.

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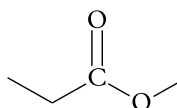
QUESTION 7

The reaction between alanine and phenylalanine is classified as

- A esterification.
- B condensation.
- C hydrolysis.
- D elimination.

QUESTION 8

The reagents required to produce the following molecule are

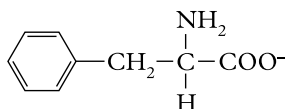


- A ethanal and methanol.
- B methanoic acid and propanol.
- C propanoic acid and methanol.
- D ethanal and ethanol.

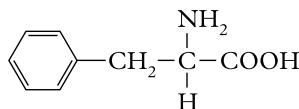
QUESTION 9

The major form of the amino acid phenylalanine in a strongly acidic solution is

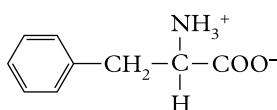
A



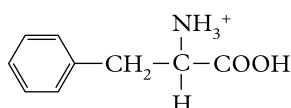
B



C



D



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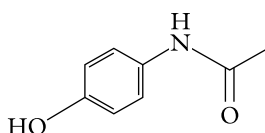
OXFORD
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In which one of the following options are all species of the set able to undergo a spontaneous redox reaction with pure water?

- A Zn, Sn^{2+} , Cl_2
- B H^+ , Ca^{2+} , Br_2
- C F_2 , Au^+ , Na
- D K, F_2 , Fe^{3+}

QUESTION 11

Paracetamol is a widely used painkiller with the structure shown below.



The functional groups in paracetamol are

- A alcohol and amine.
- B alcohol and peptide.
- C alcohol and amide.
- D ketone and amide.

QUESTION 12

Human insulin is composed of 51 amino acids. The expected number of peptide bonds in insulin is

- A 50
- B 51
- C 52
- D 102

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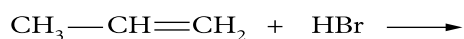
OXFORD
UNIVERSITY PRESS**QUESTION 13**

Which of the following ranks the boiling points from lowest to highest?

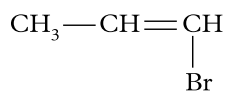
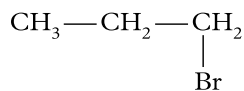
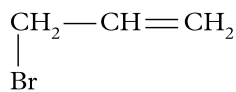
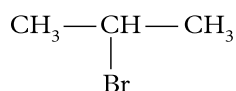
- A** Methane < methanol < methanoic acid
- B** Methane < methanoic acid < methanol
- C** Methane < ethene < propane
- D** Propanol < ethanol < methanol

QUESTION 14

Propene reacts with hydrobromic acid:



According to Markovnikov's rule, the major product that will be formed in this reaction is

A**B****C****D**

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QUESTION 15

The possible half-cell reactions in the electrolysis of hydrogen iodide (HI) are listed below:



The strongest reductant and oxidant, respectively, in this case, are

- A H^+ , H_2O
- B I^- , H^+
- C H_2O , H^+
- D H^+ , I^-

QUESTION 16

A chemist performs a titration by slowly adding a strong acid (e.g. HCl) to a weak base (e.g. NH_3). Which of the following statements correctly describes the appearance of the resulting pH titration curve?

- A The starting pH will be close to 7, and the equivalence point will be close to 7.
- B The starting pH will be low, and the equivalence point will be above 7.
- C The starting pH will be high, and the equivalence point will be below 7.
- D The starting pH will be high, and the equivalence point will be above 7.

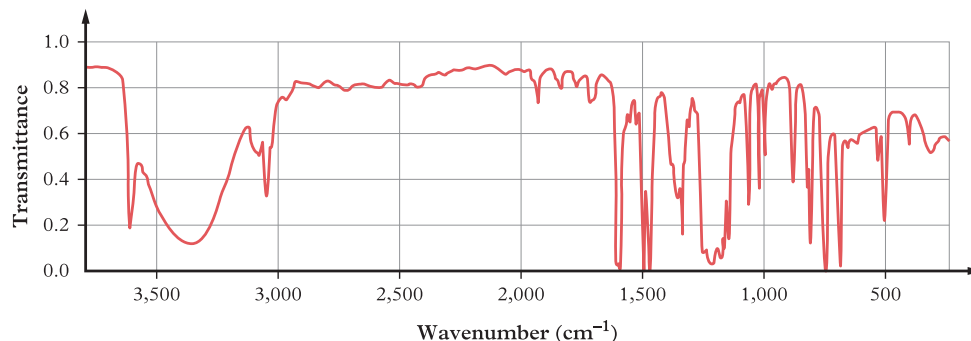
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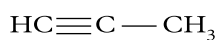
QUESTION 17

A student records the following infrared spectrum.

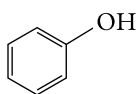


Which is the most likely compound to have this spectrum?

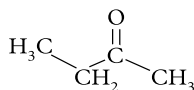
A



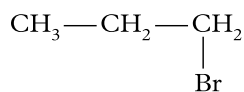
B



C



D



QUESTION 18

How many hydrogen atoms are there in a molecule of octan-3-ol?

A 8

B 18

C 19

D 20

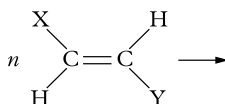
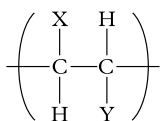
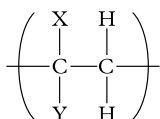
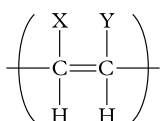
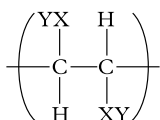
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QUESTION 19

When the following monomer undergoes addition polymerisation, the structure of the polymer formed will be

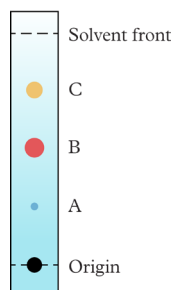
**A****B****C****D**

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QUESTION 20

A thin-layer chromatogram was obtained using the non-polar solvent hexane and polar stationary phase silica gel.



Which of the statements is correct?

- A Compound C is the most polar.
- B Compound B has a higher R_f value than compound A.
- C Compound A is the most non-polar.
- D Compound A has a higher R_f value than compound C.

Section 2: Short response

Instructions

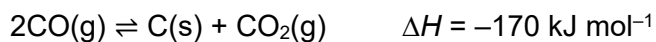
- Write in black or blue pen.
- If you need more space for a response, use the additional pages at the back of the book.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. “See page ...”
 - If you do not do this, your original response will be marked.
- This section has **7 questions** and is worth **42 marks**.

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Decomposition of carbon monoxide (CO) occurs at 550°C, to afford carbon and carbon dioxide according to the equation:



- a** The activation energy for the forward reaction is +50 kJ mol⁻¹ and the energy of CO is -220 kJ mol⁻¹. For the reverse reaction, identify and label:
- i** the energies of the reactants and products [2 marks]
 - ii** an uncatalysed reaction profile [1 mark]
 - iii** a possible profile of a catalysed reaction. [1 mark]

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b Determine the ΔH value for the following reaction: $2\text{C(s)} + 2\text{CO}_2\text{(g)} \rightleftharpoons 4\text{CO(g)}$ [1 mark]

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**QUESTION 22 (7 marks)**

Sulfuric acid (H_2SO_4) is a diprotic acid produced on an industrial scale by the contact process.

- a** Explain what is meant by the term “diprotic acid”. [1 mark]

- b** Determine the dissociation reactions of sulfuric acid in water. [2 marks]

- c** The acid dissociation constant for sulfuric acid is close to 1. Determine the expression for the acid dissociation constant for the first step and explain what this value means. [2 marks]

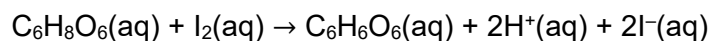
- d** Using a diagram, explain the difference between concentrated and dilute sulfuric acid. [2 marks]

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**QUESTION 23 (6 marks)**

Two students from New Zealand conducted an experiment to determine the amount of vitamin C in Ribena. They diluted 5.00 mL of Ribena to 50.0 mL in a volumetric flask. A 10 mL sample of this was taken to titrate vitamin C with iodine (0.250 M) and using starch as an indicator, according to the equation:



The following titres were obtained: 26.40 mL, 26.50 mL, 26.85 mL, 26.45 mL.

- a** Calculate the average titre you would use to calculate the concentration of vitamin C solution. [1 mark]

- b** Calculate the amount, in moles, of molecular iodine that reacted with the Ribena solution. [1 mark]

- c** Calculate the amount, in moles, of vitamin C present in 5 mL of Ribena solution. [2 marks]

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- d** Calculate the mass of vitamin C present in 5 mL of Ribena solution. [2 marks]

QUESTION 24 (4 marks)

The amino acids proline and alanine undergo condensation to form a dipeptide.

- a** Draw the structure of the product and by-product that is formed. [2 marks]

- b** Circle the peptide bond. [1 mark]

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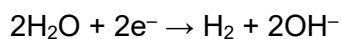
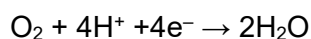
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- c Draw the structure of the dipeptide as it would exist as a zwitterion. [1 mark]

QUESTION 25 (5 marks)

Several half-cell reactions are possible in the electrolysis of hydrogen iodide, as listed below:



- a From the list above, identify the weakest reductant and oxidant. [1 mark]

- b Identify the reaction that will occur at the cathode. [1 mark]

- c Identify the reaction that will occur at the anode. [1 mark]

- d Determine the overall cell equation. [1 mark]

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- e Explain the main difference between galvanic and electrolytic cells in terms of energy transformation. [1 mark]

QUESTION 26 (7 marks)

In an electrophoresis experiment, amino acids are separated based on their net charge, which is determined by the pH of the buffer solution relative to the amino acid's isoelectric point (pI). The pI values for three common amino acids are:

Amino acid	Isoelectric point (pI)
Glycine	6.0
Lysine	9.7
Glutamic acid	3.2

- a Identify the net charge of an amino acid when it is in a buffer solution at a pH equal to its isoelectric point (pI). [1 mark]

- b Explain the net charge of glutamic acid when it is placed in a buffer at pH = 7.0. [2 marks]

- c A mixture of glycine and lysine is separated in a pH = 7.0 buffer. Determine the direction of migration for each amino acid when an electric field is applied (i.e. towards the positive or negative electrode). [2 marks]

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d Explain why lysine migrates further than glycine in the pH = 7.0 buffer. [2 marks]

QUESTION 27 (8 marks)

Five separate experiments were conducted in which pairs of reagents were mixed. This is shown in the table below. Using the electrochemical series, predict whether a spontaneous redox reaction will occur in each case. For those predicted to occur, determine the balanced chemical equation.

Reactants	Predicted reaction (yes or no)	Equation
$\text{SnCl}_2(\text{aq})/\text{Cu}$		
$\text{Fe}(\text{NO}_3)_2(\text{aq})/\text{Cl}_2(\text{g})$		
$\text{Ag}(\text{s})/\text{Cd}^{2+}(\text{aq})$		
$\text{Fe}^{3+}(\text{aq})/\text{Fe}(\text{s})$		
$\text{H}_2\text{O}(\text{l})/\text{Li}(\text{s})$		

END OF PAPER

